Tejas Bharambe

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Professional Summary

Backend Software Engineer with production experience building real-time, event-driven microservices in Python.Built the PoC solo and scaled a greenfield voice AI platform at HSBC handling **1200+ concurrent audio streams** with sub-second latency. Deep hands-on experience with **FastAPI, WebSockets, async programming, Kafka, and event-driven real-time streaming architectures**.

Work Experience

Software Engineer Jul 2025 – Present
HSBC Technology India Pune, Maharashtra

- Built the **PoC solo** and worked on a greenfield real-time call transcription service in Python and FastAPI, replacing a third-party vendor tool with an estimated potential saving of **$5M+/year** for HSBC's contact centre operations.
- Designed a **WebRTC Voice Activity Detection (VAD) pipeline** that improved ASR transcription accuracy from **45% to 92%** and reduced model hallucinations by **95%** by intelligently segmenting audio and eliminating redundant inference calls.
- **Built event-driven real-time microservices pipeline:** connection lifecycle management (1:1 per entity), frame-order guarantees via WebSocket sequencing, and delta-based Kafka publishing, sustaining 1200+ concurrent streams at sub-second latency.
- Collaborated with a sister team to **define binary payload schemas** and implemented **Kafka-based transcript publishing**, enabling real-time downstream consumption by contact centre analytics systems.
- Developed **custom load testing scripts** to simulate sustained call traffic; benchmarked **sub-second latency** and a Real-Time Factor (RTF) of ∼0.3 (3× faster than real time) under load.
- Enforced code quality using **SonarQube**, followed **CI/CD** deployment practices on **AWS**, conducted **knowledge transfer sessions**, and **mentored 2 interns**, accelerating team ramp-up.

Skills

Languages: Python, C++.

Backend: FastAPI, REST APIs, WebSockets, Asyncio, Kafka, Microservices, System Design, AWS (EC2, S3, ECS), Docker, Multiprocessing, Performance Testing.

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, SQL.

Tools: Git, GitHub, SonarQube, Jira, CI/CD.

AI-Assisted Development: GitHub Copilot, Claude Code.

Projects

Fractal Image Compression (Python, Multiprocessing, Parallel Computing) — GitHub

- Implemented a fractal image compression algorithm in Python for grayscale and RGB images. Profiled the encoding bottleneck and parallelized the pipeline across all CPU cores using Python's multiprocessing module, reducing encoding time by **50%**.

Retinal Disease Classification (PyTorch, CNN, Transfer Learning) — GitHub

- Trained a CNN using PyTorch with ResNet-based transfer learning for automated retinal disease detection, achieving **95.9% accuracy**, a **31.9% improvement** over the untrained baseline. Evaluated using confusion matrices and per-class classification reports.

Education

B.Tech, Artificial Intelligence and Data Science May 2025
Vishwakarma Institute of Technology, Pune
CGPA: 8.75/10.00

Activities and Leadership

- **Technical Executive Member**, IEEE Student Branch, VIT Pune — led data-driven PR strategy and stakeholder engagement for technical events with **500+ attendees**; managed corporate sponsorship acquisition across industry partners.